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ORIGINAL ARTICLE

Genome-wide in silico identification of LysM-RLK genes in potato (*Solanum tuberosum* L.)

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Abstract

The receptor like kinases (RLKs) belong to the RLK/Pelle superfamily, one of the largest gene families in plants. RLKs play an important role in plant development, as well as in response to biotic and abiotic stresses. The lysine motif receptor like kinases (LysM-RLKs) are a subfamily of RLKs containing at least one lysine motif (LysM) that are involved in the perception of elicitors or pathogen-associated molecular patterns (PAMPs). In the present study, 77 putative RLKs genes and three receptor like proteins were identified in potato (*Solanum tuberosum*) genome, following a genome-wide search. The 77 potato RLK proteins are classified into two major phylogenetic groups based on their kinase domain amino acid sequence similarities. Out of 77 RLKs, 10 proteins had at least one LysM. Among them three RLP proteins were found in potato genome with either 2 or three tandem LysM but these lacked a cytoplasmic kinase domain. Expression analyses of a potato LysM-RLKs (*StLysM-RLK05*) was carried out by a Real time RT-PCR, following inoculation of potato leaves and immature tubers with late blight and common scab pathogens, respectively. The expression was significantly higher in resistant than in susceptible following *S. scabies* inoculation. The *StLysM-RLK05* sequence was verified and it was polymorphic in scab susceptible cultivar. The present study provides an overview of the *StLysM-RLKs* gene family in potato genome. This information is helpful for future functional analysis of such an important protein family, in Solanaceae species.

Keywords LysM-RLK · CERK · Receptors · Plant disease resistance · Potato

Abbreviations

MAPK	Mitogen activated protein kinases
PAMPs	Pathogen-associated molecular patterns
RKs	Receptor kinases
RLKs	Receptor-like kinases
CEBiP	Chitin-elicitor binding protein
AUDPC	Area under the disease progress curve
PDB	Protein Data Bank

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Introduction

Plants have only the innate immune system, unlike animals which have both innate and adaptive immune systems [1]. Plants perceive the pathogen produced elicitors (also known as pathogen-associated molecular patterns (PAMPs)) and effectors and trigger hierarchies of downstream genes to biosynthesize the resistance related (RR) proteins and metabolites, which in turn suppress the pathogen progress in plants [2]. Bacterial cell walls contain peptidoglycans, while the fungal cell walls contain chitin, and these are perceived by plants, which in turn trigger the plant innate immune response [3]. The elicitors produced by the pathogen are recognized by the plant membrane located receptors, which in turn regulate the mitogen activated protein kinases (MAPK) leading to their phosphorylation [4].

The plant innate immune system is equipped with an array of interacting proteins, among them are the proteins involved in environmental ligand detection known as receptor kinases (RKs). The plant receptor kinases usually interact with an extracellular ligand molecule, but the cognate ligands for most of these proteins remain to be identified.

These are collectively known as receptor-like kinases (RLKs) [5]. The domain structure of a typical RLK has an N-terminal signal peptide sequence and an extracellular ligand-binding domain, followed by a transmembrane domain and an intracellular kinase domain (cytoplasmic or intracellular domain). All RLKs are targeted to plasma membrane by a 19–31 N-terminal amino acid signal peptide [6]. RLKs belong to a relatively large gene family known as "RLK/Pelle" [7, 8]. Furthermore, the domain organization and the sequence identity of extracellular domains of RLKs vary greatly among RLK gene family members [7]. The RLK/Pelle family members are subdivided into 45 subfamilies, including LysM, C-lectin and LRR subfamilies [7]. The extracellular/ligand-binding domain part of RLKs seem to vary among RLKs, the most common being the leucine-rich repeats (LRRs) and LysM containing proteins [9]. Such LRR and LysM domains are involved in diverse signaling processes such as host immunity [10], hormone signaling [11], and growth and development regulation [12]. Other than the LysM-RLKs, plants also have receptor like proteins (RLPs), which lack a cytoplasmic kinase domain [13]. RLPs resemble the extracellular domains of RLKs and seem to be involved in elicitor/stimuli perception.

Evolutionary comparative characterization of plant LysM containing protein families show that plant genomes harbor a minimum of 11 distinct types of LysM motifs [14]. The LysM motifs in plant LysM-RLKs proteins are highly diversified. At least a minimum of six distinct types of LysM motifs in plant LysM kinase proteins and five additional types of LysM motifs in non-kinase plant LysM proteins, have been identified [14]. The LysM domain which is outside the cell membrane can bind molecules like peptidoglycan and chitin along with its building blocks, *N*-acetyleglucosamine, in the case of bacteria and fungi, respectively [13]. LysM is conserved across kingdoms which share homology with the bacterial receptors [15, 16]. The LysM has a conserved secondary structure with two α helices sandwiched between two β sheets (βααβ topology) [16].

The fungal cell wall components such as chitin is being detected by cell surface receptors [17]. Despite the fact, the rice chitin-elicitor binding protein (CEBiP) lacks a functional intracellular kinase domain [18], it interacts with the rice chitin-elicitor receptor kinase 1 (OsCERK1) and trigger the down-stream genes. Nonetheless, the point mutations in *AtCERK1*(*A138H*) involving chitin biding residues did not block chitin-induced phosphorylation and initiation of an immune trigged response [19]. These findings together with the fact that a triple Atlym1/Atlym2/Atlym3 knockout mutant in *Arabidopsis thaliana*, were not able to fully respond to chitin [20], suggesting that there might be at least two chitin-activated response pathways in plants and accordingly the LysM is not required for CERK-mediated chitin-triggered defense responses. However, the recent findings suggest that CERK mediated signal perception is not fully understood in *Arabidopsis* and more evidences are required [13]. Some recent reports suggest that *AtCERK* is also reported to be involved in dual function, which is abiotic and biotic stress signalling [21]. Also, the CERK is associated with symbiosis between arbuscular mycorrhiza and other known symbiotic relationships [22]. Chitin perception by plants triggered a hierarchy of downstream genes, eventually biosynthesizing Resistance Related (RR) metabolites to suppress the pathogen spread within the plant [23]. For instance, following *Fusarium graminearum* infection of barley spikelets, many downstream regulatory genes such as *HvCERK1*, MAP kinase 3 (*HvMPK3*), MAPK substrate 1 (*HvMKS1*), several transcription factors, such as, *HvERF1/5*, *HvNAC42*, *HvWRKY23* and *HvWRKY70* were highly upregulated increasing the production of RR metabolites [23].

MAPK signaling pathway plays an important role in plant response to environmental challenges, transferring the extracellular stimuli signals into intracellular responses. However, the poor information about RLKs gene family in potato makes it very challenging to apply them in potato breeding. In the present study, we have searched the potato genome databases to identify the members of potato RLKs gene family. A comparative bioinformatics and molecular analysis of putative RLKs genes were done to elucidate the most likely RLKs involved in potato response to pathogen infections.

Materials and methods

Plant materials

For late blight experiment, a resistant potato genotype F06037 and a susceptible commercial cultivar Russet Burbank (obtained from Mrs. Agnes Murphy, Agriculture and Agri-Food Canada, NB, Canada) were used in this study. For scab experiment, a moderately resistant cultivar Russet Burbank [24] and a susceptible cultivar AG704.10 (provided by Mr. Andre Gagnon, Progest2001 Inc., QC, Canada) were used. Pre-sprouted small potato tubers were planted in pots containing promix BX and perlite at 20 ± 3 °C in the greenhouse with a 12–14 h photoperiod and $70 \pm 10\%$ relative humidity. Potato plants were fertilised once at planting with 15 g of Nutrifert (20:20:10, N:P:K).

Pathogen production, inoculation and incubation

An A2 matting type isolate of *Phytophthora infestans* (Provided by Dr. R. Peters, AAFC) was maintained on pea broth and rye B agar. This experiment was designed as a randomized block with two genotypes (resistant F06037 and susceptible Russet Burbank, two inoculations (mock and pathogen) and three replicates as described previously by

(Yogendra et al. [25]). Briefly, the fully expanded leaves, in five-week old potato plants, were inoculated on the lower surface, two spots/leaf, with 10 µl of either distilled water (Mock inoculation) or *P. infestans* sporangia (1×10^5), using a Hamilton syringe fitted with an auto dispenser (Gastight 1750DADW/S, Hamilton, Reno, NV, USA). To prevent the spread of spores, the inoculated spore suspension drops were covered with Whatman filter paper discs (4–5 mm). Finally, the inoculated plants were covered with moist transparent plastic bags to maintain high moisture to facilitate pathogen infection [26]. The plastic covers were removed 48 h post inoculation (hpi). The inoculated leaflets were harvested at 48 hpi, and disc containing lesion was cut using a 1 cm cork borer and used for gene sequencing and expression analyses. The lesion diameter was measures at 3, 6 and 9dpi, from which the area under the disease progress curve (AUDPC) was calculated [27].

The *Streptomyces scabies* (isolate EF35 was provided by Dr. C. Beaulieu, University of Sherbrook, QC) was cultured on a solid International *Streptomyces* Project 2 (ISP2) medium [28] at 28 °C for 2 weeks. This experiment was designed as randomized block, with two genotypes (resistant Russet Burbank and susceptible AG704.10), two inoculations (mock and pathogen) and three replicates. Using a cork borer, agar blocks with visible *S. scabies* growth were scooped out, placed on young tubers and incubated at 25 °C for 5 days. The diameter of diseased lesions with dark colored sunken areas or corky raised areas were recorded at 3, 6 and 9 dpi, from which the area under the disease progress curve (AUDPC) was calculated. The samples harvested at 48 hpi were used for gene expression analysis.

RLKs, LysM-RLKs and RLPs sequences and database

Three different approaches were used to retrieve RLKs, LysM-RLKs and RLPs. Initially, two fully characterized *Arabidopsis* and rice RLKs, chitin elicitor receptor kinase (CERK) were used as queries to identify homologous proteins from potato by performing a BLASTP search at PHYTOZOME v8.0 database (www.phytozome.net/) using default parameters [29] and with a cutoff *e* value of 1e−5. All splice variants were removed, and for each record, only the primary transcript was kept. Second, potato Spud DB (http://solanaceae.plantbiology.msu.edu/index.shtml) database was used to download RLKs and LysM-RLKs sequences, using *OsCERK1* gene [30] as the reference protein to identify the true orthologs in potato genome. The search was carried out using tBLASTx and the criteria considered as described by (Dhaliwal et al. [31]). Finally, the potato genome database was searched using the keywords 'serine/threonine receptor kinase', and 'Lysine motif''. The Pfam database (http://pfam.sanger.ac.uk/) was searched against the PHYTOZOME database of potato by using the HMM profiles of the LysM domain.

The Kinase domain of all LysM-RLKs and the extra cytoplasmic domain of RLPs sequences were analysed using ScanProsite program from Expasy server (http://prosite.expasy.org/).

Nomenclature, chromosomal location and gene structure

For nomenclature, prefix 'St' for *Solanum tuberosum* was added followed by RLK or RLP and numbered according to the gene position from top to bottom on the twelve potato chromosomes. The chromosomal location of each putative RLKs and RLP identified from potato database on potato chromosomes was identified using the potato genome browser at the PGSC database by BLASTP search. The identified genes were plotted onto the 12 potato chromosomes according to their ascending order of physical position and displayed using MapChart software [32].

Phylogenetic analyses

The predicted amino acid sequences of all putative potato RLKs domain were used to construct a phylogeny tree. Multiple sequence alignment (MSA) was performed by ClustalW2 and Protein Weight Matrix BLUSUM62 and default MSA Parameters. MEGA7 software was used to plot the phylogeny trees [33]. The evolutionary history of potato LysM-RLKs was inferred by using the Neighbour Joining (NJ) method with 1000 bootstrap replicates.

Quantitative reverse transcriptase PCR (qRT-PCR)

The total RNA was extracted from infected and mock leaves and tubers 48 hpi and 5 dpi, respectively, using an RNase easy kit (Qiagen, USA according to the manufacturer's instructions. The qRT-PCR reaction was carried out using specific selected RLK primers (St1331F:5′-TCGCCGGGACAAGGATTAAC-3′; St1331R:5′-TCCACTGTGCAGTCGTCAAA-3′). Almost 20 ng of cDNA in a final volume of 10 µL was used for real time PCR reaction using IQ SYBR Green Supermix (BioRad, Hercules, CA, USA) in a CFX384 Real-Time System (BioRad, ON, Canada) according to the manufacturer's instructions. The relative gene expression level to Elongation factor $1-\alpha$ ($ef-1\alpha$) as the recommended housekeeping gene was calculated for selected RLK gene according to $2^{-\Delta\Delta Ct}$ method [34].

Homology modelling of *LysM-RLK05* protein

The mRNA from two potato genotypes (Russet Burbank and AG704.10) was sequenced using specific primers.

The complete nucleotide sequence obtained from Sanger sequencing was translated into protein sequence using the ExPasy translation tool (https://web.expasy.org/translate/). The standard protein code was used and the other parameters were as default.

The obtained protein sequence was used for the homology-based modelling. HHpred was used for the homology search using 3 databases, PDB, Pfam and NCBI conserved domain [35]. The 10 best obtained proteins were used to generate the multiple sequence alignment. The obtained MSA was then used to generate the protein model using the MODELLER software [36] embedded with the MPI bioinformatics toolkit (https://toolkit.tuebingen.mpg.de/#/).

Protein–substrate binding study

The obtained protein structures were superimposed using the PyMol® tool to see the structural differences. The obtain structure was then used for docking the substrate molecules to find the differences in the substrate binding affinities. Chitin and *N*-acetylglucosamine molecular structure was downloaded from the ZINC database (http://zinc.docking.org/) [37].

The target protein, the ligand to be docked, and the docking grid, was set up using AutoDockTools [38], which is distributed as part of the MGLTools package. AutoDockTools-1.5.6. was used which is a graphical interface program that allows visualization and manipulation of molecular structures. Hydrogen atoms were reassigned to keep the polar hydrogen atoms. The orientation of the docking grid around the binding site of interest was defined and the Spacing (angstrom) factor was set to 1.000 to have the measurements taken in Angstrom units. Multiple docking runs was done by running a batch AutoDock Vina [39] via command line by using a script. The script was customized for the molecular system in terms of grid box description and the names for the input and output files.

Results and discussion

Genome-wide Identification and nomenclature of RLKs members from *Solanum tuberosum*

Potato is one of the most important food crops in the world which is challenged with many devastating pathogens. As far as our knowledge is concerned, only a few potato gene families have been identified and characterized in silico [40–46]. Since complete potato genome sequences have been made available, it was possible to identify all the RLKs family members. Therefore, the HMM and BLAST searches were performed to identify the putative RLKs proteins in potato genome. The known *A. thaliana* and *Oryza sativa* RLKs encoding genes were used as queries. All sequences with low reliable similarities (e-value cut-off of $1e^{-3}$) were removed. The remaining genes/proteins were further inspected for a significant match to the well annotated LysM domain (LysM: PF01476; E value\0.001) using the Pfam database (http://pfam.xfam.org/). Furthermore, a BLAST search of refined sequences followed by a carful manual editing to remove the redundant sequences revealed that the potato genome has 77 putative RLKs genes including five already known RLKs (Table 1). Out of 77 RLKs genes identified, 10 genes encoded a LysM containing receptor protein, that is LysM-RLKs protein. One or two LysM domains were identified in these LysM-RLKs proteins. Three proteins did not have a cytoplasmic kinase domain, belonging to receptor like proteins (RLP) family were also identified (Table 2). The proteins lacking the kinase domain had either two or three LysM domains in tandem repeats (Fig. 1). More than 90% of RLKs lack an extra cytoplasmic domain (Fig. 1), suggesting no apparent role of RLKs in the perception of PAMPs.

Interestingly, most of the LysM-RLKs existence was supported by EST hits and full-length cDNA sequences from GenBank database (Table 2) except for 4 LysM-RLKs where no such data were available which may indicate either all four LysM-RLKs are not expressed in the conditions used for analysis or that they are expressed in very low quantity. A keyword search against the NCBI, UniProt and EnsemblPlants databases resulted in identification of only one previously annotated potato LysM-RLKs sequences, namely StRLK014 (UniprotKB: Q84XG7). Since the annotations of potato RLKs proteins reported in the UniProtKB, NCBI and potato Spud DB database were highly ambiguous and uninformative, an uniform nomenclature was assigned to 77 potato RLK and three RLP proteins so that the identified potato RLK, LysM-RLK and RLP proteins can be designated as StRLK, LysM-RLK and StRLP, followed by a number 1–67, 1–10 and 1–3 respectively, based on the position of their corresponding genes on chromosomes.

Modular structure and chromosomal location of LysM-RLK and RLKs

Bioinformatic comparative analysis revealed that RLKs and RLPs possess a modular structure (Fig. 1). A typical RLK consist of an N-terminal ligand binding extracellular domain, a single membrane spanning region, and an intracellular C-terminal kinase domain involved in downstream signal transduction (Fig. 1). However, out of 77 putative potato RLKs, seventeen proteins did not have a signal peptide, 9 did not have a transmembrane domain, whereas 8 genes neither had a signal peptide nor a transmembrane domain. Four genes possess two transmembrane domains,

Table 1 Potato (*Solanum tuberosum* L.) RLK genes investigated in this study with their corresponding allocated names, PGSC IDs and protein length, Chromosome positions

PGSC protein ID	PGSC gene ID	PGSC transcript ID	Name	#	Start	End	#A.A
PGSC0003DMP400037016	PGSC0003DMG400021353	PGSC0003DMT400055038	StRLK01	1	3,085,317	3,089,347	955
PGSC0003DMP400041518	PGSC0003DMG400024010	PGSC0003DMT400061697	StRLK02	1	18,745,241	18,755,697	386
PGSC0003DMP400015919	PGSC0003DMG400009047	PGSC0003DMT400023349	StRLK03	1	62,812,198	62,822,152	731
PGSC0003DMP400049736	PGSC0003DMG400028561	PGSC0003DMT400073506	StRLK04	1	76,076,328	76,081,627	449
PGSC0003DMP400049835	PGSC0003DMG400028604	PGSC0003DMT400073645	StRLK05	1	76,405,999	76,409,982	433
PGSC0003DMP400031789	PGSC0003DMG400018248	PGSC0003DMT400046983	StRLK06	1	78,752,942	78,758,519	508
PGSC0003DMP400031817	PGSC0003DMG400018263	PGSC0003DMT400047023	StRLK07	1	79,384,073	79,392,204	947
PGSC0003DMP400044667	PGSC0003DMG400025784	PGSC0003DMT400066269	StRLK08	1	84,389,893	84,394,548	635
PGSC0003DMP400044974	PGSC0003DMG400025921	PGSC0003DMT400066706	StRLK09	1	85,214,987	85,219,870	942
PGSC0003DMP400001376	PGSC0003DMG400000703	PGSC0003DMT400001868	StRLK10	1	85,850,236	85,854,803	682
PGSC0003DMP400032471	PGSC0003DMG400018637	PGSC0003DMT400047957	StRLP01	1	87,862,741	87,865,242	353
PGSC0003DMP400002170	PGSC0003DMG400001185	PGSC0003DMT400003006	StRLK11	2	24,620,485	24,622,699	617
PGSC0003DMP400049536	PGSC0003DMG400028454	PGSC0003DMT400073238	StRLK12	2	31,320,777	31,328,950	772
PGSC0003DMP400038906	PGSC0003DMG400022436	PGSC0003DMT400057792	StRLK13	2	34,207,523	34,212,424	1716
PGSC0003DMP400028741	PGSC0003DMG400016433	PGSC0003DMT400042369	StRLK14	2	36,716,456	36,720,973	622
PGSC0003DMP400026200	PGSC0003DMG400014892	PGSC0003DMT400038554	StRLK15	2	40,623,292	40,627,543	494
PGSC0003DMP400061022	PGSC0003DMG400038918	PGSC0003DMT400089347	StLysM-RLK01	2	44,800,657	44,802,417	586
PGSC0003DMP400069450	PGSC0003DMG400047346	PGSC0003DMT400097775	StLysM-RLK02	2	44,803,921	44,805,846	641
PGSC0003DMP400017893	PGSC0003DMG400010093	PGSC0003DMT400026193	StRLK16	2	44,946,734	44,953,393	682
PGSC0003DMP400002669	PGSC0003DMG400001470	PGSC0003DMT400003734	StRLK17	2	45,266,248	45,271,733	422
PGSC0003DMP400035096	PGSC0003DMG400020216	PGSC0003DMT400052075	StLysM-RLK03	2	47,881,442	47,883,531	669
PGSC0003DMP400035179	PGSC0003DMG400020242	PGSC0003DMT400052190	StRLK18	2	48,147,151	48,150,472	449
PGSC0003DMP400004510	PGSC0003DMG401002542	PGSC0003DMT400006515	StRLK19	3	60,098,538	60,105,967	588
PGSC0003DMP400022704	PGSC0003DMG400012821	PGSC0003DMT400033378	StRLK20	3	1,981,508	1,991,635	370
PGSC0003DMP400031066	PGSC0003DMG400017779	PGSC0003DMT400045838	StRLK21	3	4,371,570	4,377,186	728
PGSC0003DMP400001272	PGSC0003DMG400000642	PGSC0003DMT400001716	StRLK22	3	56,635,358	56,640,706	367
PGSC0003DMP400024973	PGSC0003DMG403014197	PGSC0003DMT400036821	StRLK23	3	57,054,280	57,061,338	445
PGSC0003DMP400009916	PGSC0003DMG400005632	PGSC0003DMT400014360	StRLP02	3	58,933,696	58,936,005	390
PGSC0003DMP400004478	PGSC0003DMG400002526	PGSC0003DMT400006475	StLysM-RLK04	3	60,494,235	60,402,303	663
PGSC0003DMP400004649	PGSC0003DMG400002602	PGSC0003DMT400006692	StRLK24	3	60,621,274	60,626,772	789
PGSC0003DMP400004627	PGSC0003DMG400002593	PGSC0003DMT400006664	StRLK25	3	60,836,966	60,744,444	737
PGSC0003DMP400004416	PGSC0003DMG400002497	PGSC0003DMT400006390	StRLK26	3	60,996,500	60,903,121	885
PGSC0003DMP400016924	PGSC0003DMG400009571	PGSC0003DMT400024762	StRLK27	4	13,860,796	13,866,198	668
PGSC0003DMP400021220	PGSC0003DMG400011992	PGSC0003DMT400031291	StRLK28	4	45,792,032	45,794,715	560
PGSC0003DMP400017642	PGSC0003DMG400009986	PGSC0003DMT400025866	StRLK29	4	71,691,994	71,695,628	370
PGSC0003DMP400043552	PGSC0003DMG400025102	PGSC0003DMT400064597	StRLK30	5	1,585,208	1,591,150	745
PGSC0003DMP400046649	PGSC0003DMG400026867	PGSC0003DMT400069067	StRLK31	5	9,559,034	9,567,505	379
PGSC0003DMP400014933	PGSC0003DMG400008506	PGSC0003DMT400021929	StRLK32	5	16,416,409	16,428,188	468
PGSC0003DMP400007826	PGSC0003DMG400004409	PGSC0003DMT400011261	StRLK33	6	5,048,260	5,051,643	401
PGSC0003DMP400034849	PGSC0003DMG402020091	PGSC0003DMT400051759	StRLK34	6	58,440,705	58,445,444	438
PGSC0003DMP400030204	PGSC0003DMG400017291	PGSC0003DMT400044535	StRLK35	7	509,106	50,912,801	385
PGSC0003DMP400047758	PGSC0003DMG400027458	PGSC0003DMT400070632	StRLK36	7	2,254,360	2,258,204	391
PGSC0003DMP400053808	PGSC0003DMG400030894	PGSC0003DMT400079361	StRLK37	7	4,189,437	4,193,398	1027
PGSC0003DMP400031150	PGSC0003DMG400017836	PGSC0003DMT400045970	StRLK38	7	35,462,547	35,471,257	402
PGSC0003DMP400001331	PGSC0003DMG402000674	PGSC0003DMT400001803	StLysM-RLK05	7	44,893,452	44,901,835	626
PGSC0003DMP400032008	PGSC0003DMG400018367	PGSC0003DMT400047295	StRLK39	7	48,601,139	48,612,113	792
PGSC0003DMP400035500	PGSC0003DMG400020438	PGSC0003DMT400052650	StRLK40	7	50,302,374	50,302,374	1089
PGSC0003DMP400021906	PGSC0003DMG400012392	PGSC0003DMT400032254	StRLK41	7	52,766,165	52,769,860	430
PGSC0003DMP400033480	PGSC0003DMG400019267	PGSC0003DMT400049600	StRLK42	7	54,119,844	54,125,226	479
PGSC0003DMP400025579	PGSC0003DMG402014514	PGSC0003DMT400037638	StRLK43	8	43,140,298	43,151,367	580
PGSC0003DMP400014906	PGSC0003DMG400008493	PGSC0003DMT400021884	StRLK44	9	165,492	168,003	403
PGSC0003DMP400015669	PGSC0003DMG400008907	PGSC0003DMT400023009	StRLK45	9	1,738,955	1,742,391	416
PGSC0003DMP400022995	PGSC0003DMG402012981	PGSC0003DMT400033789	StRLK46	9	4,352,151	4,362,118	630
PGSC0003DMP400003110	PGSC0003DMG400001732	PGSC0003DMT400004363	StRLK47	9	6,573,232	6,582,777	928

Table 1 (continued)

PGSC protein ID	PGSC gene ID	PGSC transcript ID	Name	#	Start	End	#A.A
PGSC0003DMP400048528	PGSC0003DMG400027905	PGSC0003DMT400071735	StRLK48	9	6,777,562	6,786,977	390
PGSC0003DMP400015350	PGSC0003DMG400008728	PGSC0003DMT400022521	StRLK49	9	31,862,437	31,866,230	942
PGSC0003DMP400030833	PGSC0003DMG400017650	PGSC0003DMT400045499	StRLK50	9	44,062,394	44,065,820	442
PGSC0003DMP400010798	PGSC0003DMG400006086	PGSC0003DMT400015590	StLysM-RLK06	9	55,222,796	55,225,151	625
PGSC0003DMP400010799	PGSC0003DMG400006087	PGSC0003DMT400015591	StLysM-RLK07	9	55,225,654	55,228,002	626
PGSC0003DMP400010800	PGSC0003DMG400006088	PGSC0003DMT400015592	StLysM-RLK08	9	55,228,761	55,233,189	628
PGSC0003DMP400035792	PGSC0003DMG400020594	PGSC0003DMT400053083	StRLK51	9	59,759,585	59,763,402	932
PGSC0003DMP400019984	PGSC0003DMG400011284	PGSC0003DMT400029366	StRLK52	10	346,059	349,069	388
PGSC0003DMP400019968	PGSC0003DMG400011274	PGSC0003DMT400029343	StRLK53	10	693,052	696,920	386
PGSC0003DMP400025340	PGSC0003DMG400014392	PGSC0003DMT400037295	StRLK54	10	1,269,043	1,272,853	396
PGSC0003DMP400030729	PGSC0003DMG400017596	PGSC0003DMT400045344	StRLK55	10	52,239,559	52,247,269	371
PGSC0003DMP400014755	PGSC0003DMG400008403	PGSC0003DMT400021662	StRLK56	10	53,211,328	53,217,628	964
PGSC0003DMP400018498	PGSC0003DMG400010460	PGSC0003DMT400027119	StRLK57	10	54,972,996	54,975,552	347
PGSC0003DMP400041017	PGSC0003DMG400023704	PGSC0003DMT400060941	StRLK58	10	57,404,196	57,407,235	366
PGSC0003DMP400049978	PGSC0003DMG400028688	PGSC0003DMT400073837	StRLK59	11	6,627,803	6,632,438	938
PGSC0003DMP400001703	PGSC0003DMG400000897	PGSC0003DMT400002350	StRLP03	11	11,066,324	11,069,711	413
PGSC0003DMP400042247	PGSC0003DMG400024426	PGSC0003DMT400062758	StRLK60	11	30,500,053	30,503,767	716
PGSC0003DMP400014242	PGSC0003DMG400008093	PGSC0003DMT400020901	StLysM-RLK09	11	41,411,928	41,413,583	551
PGSC0003DMP400014236	PGSC0003DMG400008090	PGSC0003DMT400020895	StRLK61	11	41,455,580	41,459,934	620
PGSC0003DMP400044002	PGSC0003DMG400025365	PGSC0003DMT400065260	StRLK62	11	44,643,403	44,648,951	463
PGSC0003DMP400026987	PGSC0003DMG400015391	PGSC0003DMT400039817	StRLK63	12	180,285	184,289	627
PGSC0003DMP400005217	PGSC0003DMG400002894	PGSC0003DMT400007503	StRLK64	12	3,071,868	3,078,021	375
PGSC0003DMP400023346	PGSC0003DMG400013199	PGSC0003DMT400034331	StRLK65	12	7,178,247	7,181,678	485
PGSC0003DMP400049857	PGSC0003DMG400028617	PGSC0003DMT400073675	StRLK66	12	50,035,429	50,041,955	924
PGSC0003DMP400056149	PGSC0003DMG402033647	PGSC0003DMT400083900	StRLK67	12	51,847,368	51,850,923	584
PGSC0003DMP400060418	PGSC0003DMG400038314	PGSC0003DMT400088743	StLysM-RLK10	12	56,788,368	56,790,564	573

whereas 17 genes had at least one transmembrane domain and a signal peptide.

The genome of potato comprises of 12 chromosomes ranged from 88.7 to 45.5 Mb corresponding to chromosome 1 and 11, respectively. Potato chromosomal localization of genes encoding putative RLKs, LysM-RLK and RLPs is shown in Fig. 2. In silico mapping of RLKs and RLPs on potato chromosomes shows an uneven distribution of the genes on all the 12 chromosomes. Chromosomal localization analysis revealed that the 77 RLK genes and 3 RLPs were found to be distributed among twelve potato chromosomes. All the twelve potato chromosomes have at least a copy of putative RLKs. Chromosome 1, 2 and 9 each with 11 and chromosome 8 with only one copy, had the highest and the lowest number of RLKs, respectively (Table 1). Furthermore, the patterns of RLKs distribution on individual potato chromosomes also revealed that certain physical regions have a relatively higher accumulation of RLKs gene clusters. For instance, RLK genes located on chromosomes appear to be congregated at the end of the chromosomes arms. In particular, LysM-RLKs are distributed on chromosome 2, 3, 7, 9, 11 and 12. Chromosomes 2 and 9 with three LysM-RLKs had the highest, whereas chromosomes 4, 5, 6 and 8 and 10 lacks even a single copy of LysM-RLK genes (Fig. 3). Genes belonging to a gene family are often distributed in clusters at certain chromosomal regions. For instance, gene families in rice, poplar and soybean were also found to be distributed in clusters [47, 48]. The potato genome seems to exhibit a relatively high plasticity, because of gene duplication events and other important structural mutations that are found in a high frequency (Consortium [49]. The total number of LysM-RLKs exons ranged from 1 to 12, whereas three RLPs had either 4 or 5 exons (data not shown).

LysM-RLKs in potato genome

The LysM-RLKs known as LYK proteins in some references have an extracellular domain possessing 1–3 LysMs, a single transmembrane domain and an intracellular Ser/Thr kinase domain. In *A. thaliana* genome 5 LysM-RLKs are reported whereas 10 LysM-RLKs were identified in this study [20]. A similar number of LysM-RLK proteins was also found in the *Medicago truncatula* (17 proteins), soybean (12 proteins) and tomato (14 proteins) genome [50, 51]. The number of LysM-RLKs genes vary greatly among members of plant kingdom. For instance, soybean and poplar genomes have a greater number of LysM-RLKs genes than *Arabidopsis*, rice and *M. truncatula* [52]. The LYK proteins have been shown to play an essential role as receptors for the perception of the lipochitooligosaccharide nodulation factors (NFs) that

Table 2 List of LysM-RLK genes in potato (*Solanum tuberosum* L.) along with their corresponding allocated names, CDS and protein length, chromosome positions and other important information

Transcript ID	Name	#Chro	CDS	Protein	#LysM	#Exons	Kinase	RNA-seq	EST data	UniProt	RefSeq
PGSC0003DMT400097775	StLysM-RLK02	2	1926	641aa	1	1	+	A	+	M1E1E1	XM_006338210.2
PGSC0003DMT400089347	StLysM-RLK01	2	1761	586aa	1	1	+	A	–	M1DHZ4	XP_006338759.1
PGSC0003DMT400088743	StLysM-RLK10	12	1722	573aa	1	2	+	A	+	M1DGM0	XP_015162146.1
PGSC0003DMT400006475	StLysM-RLK04	3	2472	663aa	1	11	+	A	+	M0ZRF0	XP_006343034.2
PGSC0003DMT400020901	StLysM-RLK09	11	1656	551aa	1	1	+	Na	–	M1AEA1	XP_006340406.2
PGSC0003DMT400015591	StLysM-RLK07	9	2177	626aa	2	2	+	A	+	M1A692	XM_006362120.2
PGSC0003DMT400001803	StLysM-RLK05	7	2588	626aa	2	12	+	A	+	M0ZJ42	XM_006357883.2
PGSC0003DMT400015592	StLysM-RLK08	9	1887	628aa	2	2	+	A	+	M1A693	XM_006362121.2
PGSC0003DMT400052075	StLysM-RLK03	2	2010	669aa	2	2	+	Na	–	M1BSR7	XP_006338804.1
PGSC0003DMT400015590	StLysM-RLK06	9	2184	625aa	2	2	+	A	+	M1A691	XM_006362119.2
PGSC0003DMT400047957	StRLP01	1	1406	353aa	2	4	–	A	+	Q84XG7	NP_001275479.1
PGSC0003DMT400002350	StRLP03	11	1653	413aa	2	5	–	A	+	M0ZJZ7	XM_006351839.2
PGSC0003DMT400014360	StRLP02	3	1397	390aa	3	5	–	Na	–	M1A431	XM_006341551.2

A available, *Na* not available

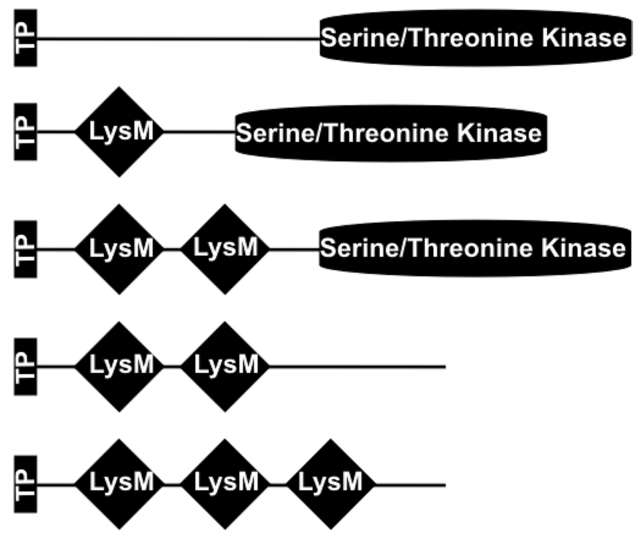


Fig. 1 The domain organization of the typical plant RLKs: A canonical RLKs containing a flexible N-terminal extracellular domain, a transmembrane domain, and a C-terminal intracellular serine/threonine kinase domain. RLKs have been grouped into three subtypes based on the sequence and structural variability of the extracellular domain. A RLP lacks a cytoplasmic kinase domain. A representative of each group is depicted in the figure; TP is transit Peptide

are produced by rhizobia as a part of nitrogen-fixing symbiotic procedure [53]. LysMs are known to bind *N*-acetylglucosamine (GlcNAc)-containing glycan molecules including, chitin, peptidoglycan, and NFs specifically recognizing only their cognate signal differ in only a few, critical amino acid residues [54]. In this study, potato LysM-RLKs had one or two LysMs (Table 2). Almost all known LysM-RLK proteins motifs contain a maximum of three LysM copies [52].

Phylogenetic relationships among the RLK genes

Potato RLKs phylogeny analysis was done using either LysM motif sequences or the full kinase domains when applicable. Interestingly, the two sets of phylogeny trees matched each other quite well (data not shown). Therefore, only the kinase domain sequence tree is represented in Fig. 4. In the phylogenetic tree two major clades representing LysM containing and without LysM motif were identified. 10 proteins were grouped in LysM containing clade and 67 were grouped in LysM lacking clade. All LysM containing proteins were grouped in one distinct clade. The LysM containing protein clade was further divided into two sub-groups consisting of RLKs with one or two LysM motifs, suggesting that the kinase domain of LysM-RLKs possess enough information to distinguish RLKs. Interestingly, StRLK04 and StRLK14 proteins with lysin motif receptor kinase (LYK) annotation were also grouped in LysM clade. Although, neither StRLK04 nor StRLK14 had a LysM according to Prosite database, they shared extensive sequence similarity to LYK

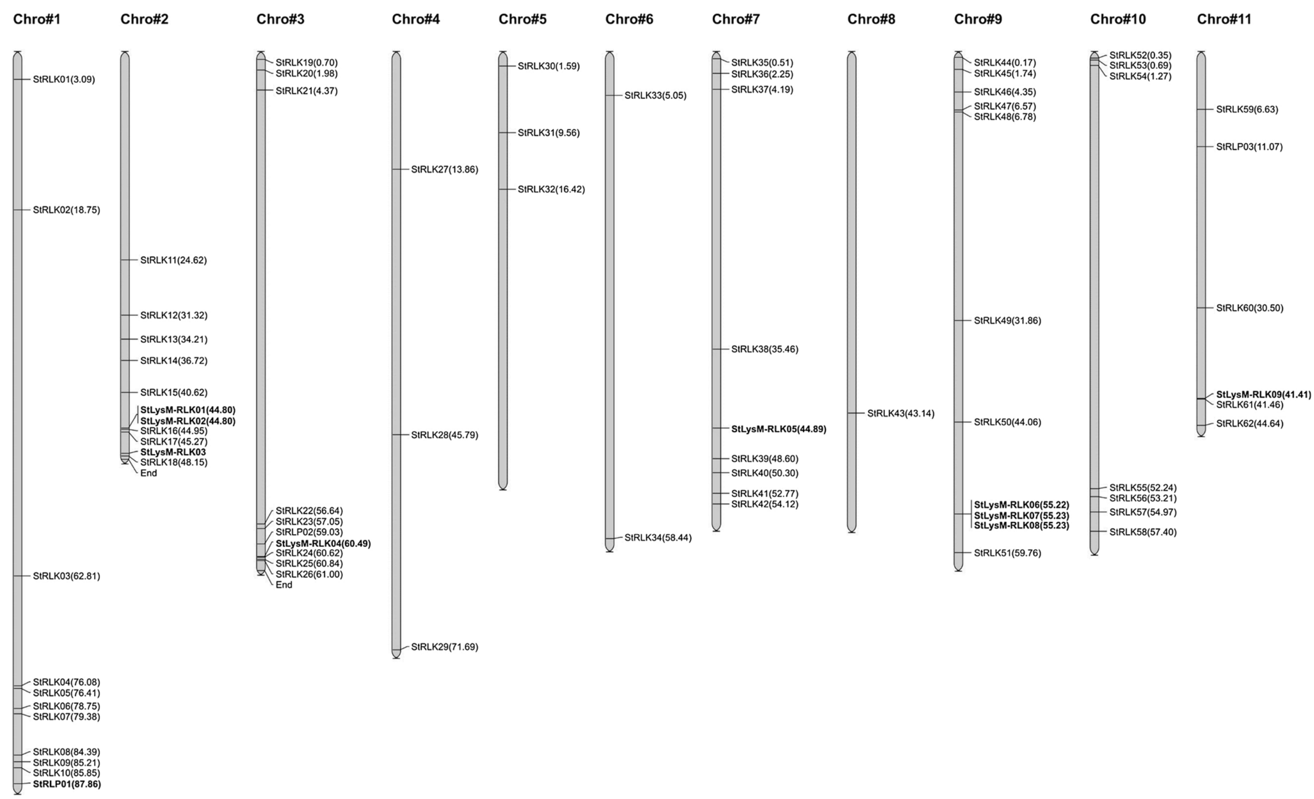
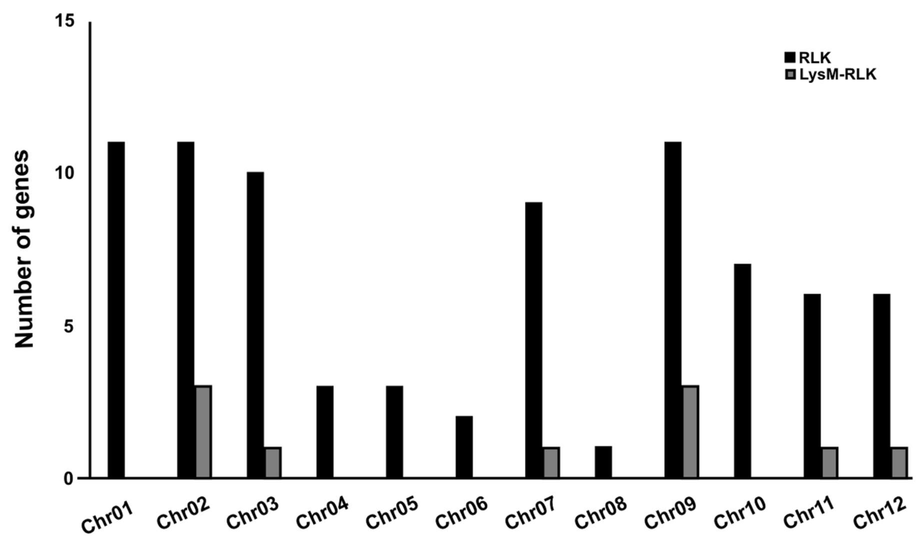


Fig. 2 Potato chromosomal localization of genes encoding putative RLKs and RLPs identified in this study. Values in parenthesis represent the position of each gene on the chromosome. Coloured boxes indicate duplicated gene clusters

Fig. 3 The distribution of potato RLKs/LysM-RLKs over 12 potato chromosomes



proteins from model plants. StRLK04 had 76.5% and 95% similarity to *A. thaliana* CERK1 (AT3G21630.1) and *S. lycopersicum* LYK13 (Solyc01g098410.2.1), respectively. Moreover, StRLK14 had 70.4% and 97.5% similarity to *A. thaliana* CERK1 (AT3G21630.1) and *S. lycopersicum* LYK11 (Solyc02g081040.2.1), respectively.

In line with other findings, the extracellular part of LysM-RLK potato proteins showed a highly variable sequence, suggesting a recognition specificity for different ligands [55, 56]. None of LysMs from potato Lys-RLK proteins shared more than 73% sequence identity, suggesting that they may have been subjected to positive selection to recognize a range of various ligands/elicitors [55].

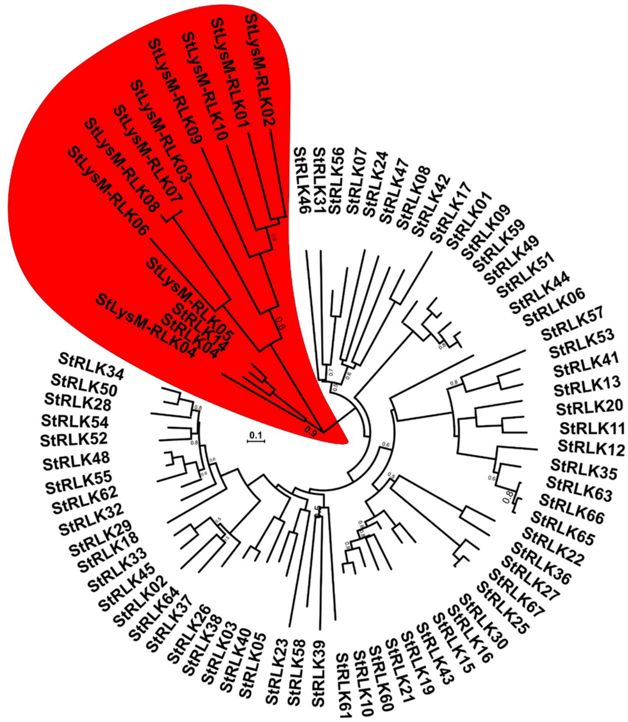


Fig. 4 An evolutionary relationship of 77 potato RLKs. The evolutionary history of RLKs was inferred by using the Neighbor-Joining method. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed by using the JTT method. The rate variation among sites was modeled with a gamma distribution. Positions with less than 95% site coverage were eliminated from the data. LysM-RLK clade of potato RLKs kinase domain was labeled with red color for simplifying the figure. The PGSC ID of all potato RLKs can be found in potato spud DB (http://solanaceae.plantbiology.msu.edu/integrated_searches.shtml)

Sequence analysis of StLysM-RLK05

Based on RNA-seq data [57] and the highest sequence similarity between potato LysM-RLK and RLPs to *A. thaliana* and *O. sativa* corresponding proteins, one LysM-RLKs (PGSC0003DMT400001803) was selected for sequencing and qRT-PCR analyses. To elucidate the selected LysM-RLKs DNA and protein sequence differences between resistance and susceptible potato genotypes, the entire selected LysM-RLK was amplified and Sanger sequenced. Multiple sequence alignment of LysM part of selected LysM-RLK from resistance and susceptible potato genotypes with that of *A. thaliana* and *S. tuberosum* Group Phureja DM1-3 reveal that despite genetic diversity at DNA level as single nucleotide polymorphisms (SNPs), no amino acid substitution occurred in the protein sequences (Fig. 5). A ten amino acid deletion outside of LysM part is seen in the doubled monoploid *S. tuberosum* Group Phureja DM1-3 genotype as compared to Russet Burbank, AG704.10 and F06037 tetraploid genotypes (Fig. 5). Although ten amino acid deletions in *S. tuberosum* Group Phureja DM1-3 genotype is relatively big, this deletion does not seem to affect the LysM-RLKs ligand binding (Fig. 5). It can be concluded that the corresponding sequence is not involved in chitin binding in CERK proteins from *Arabidopsis* and rice [22, 30].

StLysM-RLK05 gene expression in resistant and susceptible genotypes

The late blight and scab severities of potato genotypes significantly varied. The late blight severity measured as lesion diameter was 55.43 mm (AUDPC = 160.77) in susceptible Russet Burbank as opposed to 26.79 mm (AUDPC = 88.85) in resistant F06037) [27]. The common scab disease measured as lesion diameter was – 10.6 mm^2 in susceptible AG704.10 as opposed to 5.8 mm^2 in resistant Russet Burbank.

To investigate the effect of late blight and common scab pathogens on selected *StLysM-RLK05* gene expression pattern, potato of leaves a susceptible (Russet Burbank) and a resistant (F06037) genotypes and potato tubers of a resistant potato genotype Russet Burbank and a susceptible commercial cultivar AG704.10 were inoculated with *P. infestans* and *S. scabiei,* respectively. The abundance of

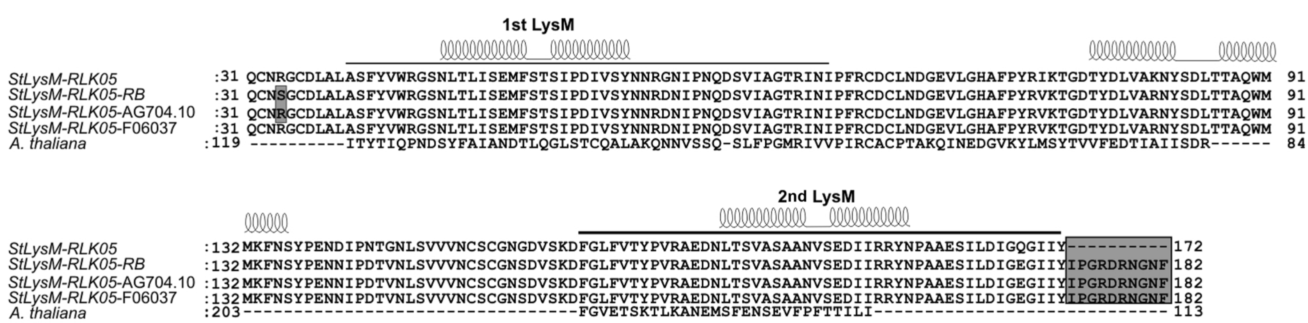


Fig. 5 Comparison of protein sequence (StLysM-RLK05) variation among Russet Burbank, AG704.10 and F06037. Box shows the deletion of ten amino acid in *S. tuberosum* Group Phureja DM1-3s LysM

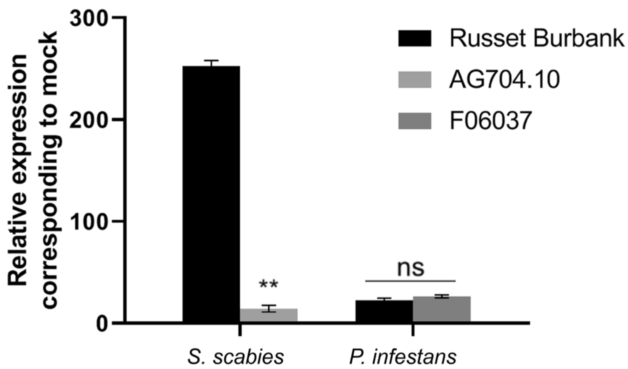


Fig. 6 The expression of *StLysM-RLK05* gene, in late blight resistant genotype F06037 and susceptible Russet Burbank, inoculated with *P. infestans* relative to mock inoculation, and in scab susceptible AG704.10 and moderately resistant Russet Burbank, inoculated with *S. scabies* relative to mock inoculated, based on qRT-PCR. The qRT-PCR data were normalized using potato elongation factor 1-a gene. Error bars indicate the standard deviation which were calculated from three independent biological replicates

StLysM-RLK05 gene transcripts significantly (P < 0.01) increased in response to *P. infestans* and *S. scabiei* treatments in both genotypes, confirming the efficacy of late blight and scab diseases on *StLysM-RLK05* gene upregulation. *S. scabies* inoculated plants showed higher expression of StLysM-RLK05 gene compared to the *P. infestans* inoculated plants. After *S. scabies* inoculation, 8 and 259-fold increase in mRNA transcript abundance of *StLysM-RLK05* gene was recorded in susceptible AG704.10 and resistant Russet Burbank, respectively (Fig. 6). The expression level of *StLysM-RLK05* increased relative to mock almost by 30-fold in both resistant F06037 susceptible Russet Burbank genotypes, at 48 h post *P. infestans* inoculation, with no significant difference (P > 0.05). The study on expression profile of important members of gene families using publicly available data such as EST and RNA-seq as well as validation of the expression pattern of selected genes using qRT-PCR is of great benefit to reveal the function of newly identified genes. Results of qRT-PCR analysis of *StLysM-RLK05* gene in leaves and tubers matched quite well with EST and RNA-seq data, strongly suggesting that preliminary expression profiling of RLKs genes using publicly available expression data followed by their validation using qRT-PCR provide more reliable expression profile of RLKs family members. In line with the findings of this study, the expression of receptor kinases was significantly increased in tomato, following inoculation with *Phytophthora parasitica* [58]. Furthermore, expression of *A. thaliana* orthologs was also found to be highly induced by pathogen infections. For instance, qRT-PCR and microarray analysis in *A. thaliana* showed that a LysM RLK1 gene expression was increased when seedling were challenged with fungus *Erysiphe cichoracearum* [59].

Homology modelling and protein–substrate docking

The amino acid sequence obtained from the ExPasy translation tool was aligned with the CERK domain from Protein Data Bank (PDB) which is presented in the Fig. 5. The difference due to the polymorphism in the mRNA sequence at the amino acid level is marked; in AG704.10 Serine (S) was changed to Arginine (R). This substitution of a polar uncharged amino acid to a non-polar charged amino acid can make a difference at the functional level.

To further investigate the difference in the function we used the protein–ligand docking studies to evaluate if there is any binding affinity difference in the substrates (Fig. 7). We used three substrates viz; chitin monomer (ZINC ID-9970239), chitohexose and *N*-acetylglucosamine (ZINC ID-3983907). We found no difference in the substrate binding affinity between Russet Burbank and AG704.10, except for *N*-acetylglucosamine. The binding affinity was lower for AG704.10 when compared to the Russet Burbank. Hence, we hypothesize that single domain even with the polymorphism doesn't make significant difference to the *LysM-RLK05* substrate binding. Furthermore, the difference between single and multiple domain proteins should be tested to find if multiple domain proteins have advantage in signal perception and transduction.

In conclusion, the current study identified 77 RLK genes and three RLP in potato genome, and analyzed their chromosomal distribution, structure, phylogeny, and expression profile of one LysM-RLK. This study provides a comprehensive information on the RLK gene family and can help researchers in better understanding the functional activities of such receptor genes in potato and further application in genome editing to enhance disease resistance [60].

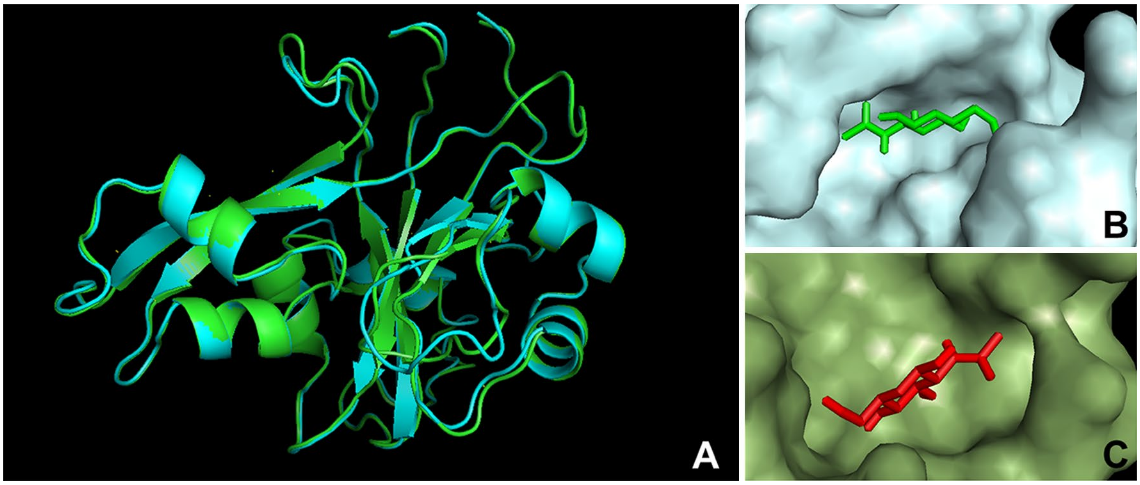


Fig. 7 Superimposed secondary structures of LysM-RLK05 protein modelled with homology modelling Green (**a**). *N*-acetylglucosamine binding to the Russet Burbank LysM-RLK05 (**b**) and *N*-acetylglucosamine binding to the AG704.10 LysM-RLK05 (**c**). The binding to *N*-acetylglucosamine in AG704.10 is shallow when compared to Russet Burbank which also coincides with the binding affinity

Author contributions FN conducted the experiments, bioinformatics analyses and wrote the manuscript; SJ conducted inoculation studies and edited the manuscript; HX conducted scab inoculation studies; AK conceived the idea and edited the manuscript.

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Compliance with ethical standards

Conflict of interest Authors declare no conflicts of interest.

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